Chapter 9: Atomic Foundations of Matter

Class 9 Science (NCERT, Exploration) | Quick revision notes

1. Laws of Chemical Combination

Law of Conservation of Mass (Lavoisier, 1789)

- Mass can neither be **created nor destroyed** in a chemical reaction: total mass of reactants = total mass of products.
- Physical changes (dissolving salt, tearing paper) also show no change in mass.
- Vinegar + baking soda: in an **open** flask the reading falls because CO_2 escapes; with a balloon fitted (**closed system**) the mass stays the same.
- Sodium sulfate + barium chloride → barium sulfate (white precipitate) + sodium chloride; mass unchanged.

Law of Constant / Definite Proportions (Proust)

- A compound always contains the same elements combined in a **fixed ratio by mass**, whatever its source or method of preparation.
- Water: H : O = **1 : 8** by mass (9 g water gives 1 g H and 8 g O). NaCl: Na : Cl = 23 : 35.5.
- Holds for compounds, **not** for mixtures (mixtures can have any ratio).

Solved idea (ratio method): 12 g C + 32 g O_2 gives 44 g CO_2. So 2.4 g C gives (44/12) × 2.4 = **8.8 g CO_2**. Likewise, 46 g Na needs (35.5/23) × 46 = **71 g Cl**.

2. Dalton's Atomic Theory (1808)

These two laws led to Dalton's postulates:

- All matter is made of very tiny particles called **atoms**.
- Atoms are **indivisible** and cannot be created or destroyed in a chemical reaction; they only rearrange.
- Atoms of a given element are **identical** in mass and chemical properties.
- Atoms of different elements have **different** masses and chemical properties.
- Atoms combine in the ratio of **simple whole numbers** to form compounds.
- The relative **number and kinds** of atoms are **constant** in a given compound.

Example: H and O atoms combine to form water, but the atoms themselves are not destroyed. When magnesium burns in air, Mg atoms combine with O atoms to form white magnesium oxide.

3. Molecules and Chemical Bonds

- **Molecule:** an electrically neutral entity of more than one atom, capable of independent existence, showing all properties of that substance. Some elements (e.g. helium) exist as single stable atoms.
- Atoms with a full valence shell (**octet**: 8 electrons, or **duplet**: 2 if K-shell is outermost) are stable. Others **share, gain or lose** electrons to get there.
- The force holding atoms together is a **chemical bond**; bonding lowers the energy of the system, making it more stable.

A. Covalent bond: sharing of electrons

Molecule	Valence e^- needed	Bond	Representation
H_2	1 each	Single	H–H
Cl_2	1 each (7 valence e^-)	Single	Cl–Cl
O_2	2 each (6 valence e^-)	Double (2 shared pairs)	O=O
N_2	3 each (5 valence e^-)	Triple (3 shared pairs)	N≡N
HCl	H needs 1, Cl needs 1	Single	H–Cl
H_2O	O needs 2, each H needs 1	2 single bonds (2 H share with 1 O)	H–O–H

Naming covalent compounds

- Use prefixes: **mono**(1), **di**(2), **tri**(3), **tetra**(4), **penta**(5), **hexa**(6). Second element ends in **-ide**.
- **Mono-** is usually omitted for the first element. Drop the last vowel of a prefix before a vowel (*monoxide, pentoxide, tetroxide*); keep 'i' (*dioxide, trioxide*).
- If H is the first element, no prefix: H_2S = hydrogen sulfide. Common names stay: H_2O (water), NH_3 (ammonia).

CO = carbon monoxide | CO_2 = carbon dioxide | CS_2 = carbon disulfide | PCl_3 = phosphorus trichloride | SF_6 = sulfur hexafluoride | N_2O_4 = dinitrogen tetroxide | N_2O_5 = dinitrogen pentoxide

B. Ionic bond: transfer of electrons

- Atom with **fewer than 4** valence electrons tends to **lose** them (forms a **cation**, positive). Atom with **more than 4** tends to **gain/share** (forms an **anion**, negative).
- Na (2,8,1) loses 1 e^- → Na^+ (11 protons, 10 electrons). Cl (2,8,7) gains 1 e^- → Cl^-. S gains 2 e^- → S^{2-}; O forms O^{2-}; Mg loses 2 e^- → Mg^{2+}.
- The **electrostatic force of attraction** between oppositely charged ions is the **ionic bond**.
- Ionic compounds form 3-D **crystal lattices** (in NaCl each Na^+ is surrounded by 6 Cl^- and vice versa), not single molecules.
- Usually **metal + non-metal**. Naming: **cation first, then anion** (sodium chloride, calcium oxide, magnesium sulfide). Polyatomic ions generally do not end in -ide.

4. Common Ions and Valencies (Table 9.1)

Valency	Cations (+)	Anions (−)
1	Na^+, Li^+, K^+, Ag^+, Cu^+ (cuprous), NH_4^+ (ammonium)	F^-, Cl^-, Br^-, I^-, OH^- (hydroxide), NO_3^- (nitrate), HCO_3^- (hydrogencarbonate)
2	Ca^{2+}, Ba^{2+}, Mg^{2+}, Zn^{2+}, Fe^{2+} (ferrous), Cu^{2+} (cupric)	O^{2-} (oxide), S^{2-} (sulfide), CO_3^{2-} (carbonate), SO_4^{2-} (sulfate)
3	Al^{3+}, Fe^{3+} (ferric)	(none in this table)

5. Writing Chemical Formulae (Criss-Cross Method)

- **Covalent:** write symbols → write valencies → criss-cross as subscripts. H + Cl → HCl; H + S → H_2S; C + Cl → CCl_4. A subscript of 1 is not written.
- **Ionic:** cation first, then anion → criss-cross only the **numbers** of the charges → reduce to the **simplest ratio**. Charges are not shown in the formula. Total + charge = total − charge.

Compound	Ions	Criss-cross	Formula
Calcium chloride	Ca^{2+}, Cl^-	Ca_1Cl_2	**$CaCl_2$**
Aluminium oxide	Al^{3+}, O^{2-}	Al_2O_3	**Al_2O_3**
Magnesium oxide	Mg^{2+}, O^{2-}	Mg_2O_2 → simplify	**MgO**
Calcium carbonate	Ca^{2+}, CO_3^{2-}	$Ca_2(CO_3)_2$ → simplify	**$CaCO_3$**
Magnesium hydroxide	Mg^{2+}, OH^-	$Mg(OH)_2$	**$Mg(OH)_2$**
Aluminium sulfate	Al^{3+}, SO_4^{2-}	$Al_2(SO_4)_3$	**$Al_2(SO_4)_3$**

Remember: use **brackets** only when two or more of the same polyatomic ion are present: $Al(OH)_3$ (not $AlOH_3$), $Ca(NO_3)_2$. No bracket for a single polyatomic ion ($CaCO_3$, Na_2SO_4).

6. Ionic vs Covalent Compounds

Property	Ionic compounds (e.g. NaCl, $CuSO_4$)	Covalent compounds (e.g. camphor, naphthalene, sugar)
Formed by	Transfer of electrons (metal + non-metal)	Sharing of electrons (usually non-metals)
Units	3-D crystal lattice of ions	Discrete molecules
Solubility	Generally soluble in **water**; insoluble in kerosene and petrol	Mostly insoluble in water; soluble in **kerosene and petrol** (sugar is a water-soluble exception)
Conductivity	Solid: **no** (ions fixed in position). In water solution: **yes** (free ions)	Generally **do not conduct** (no ions); sugar solution also does not conduct
Melting / boiling point	**High** (strong inter-ionic attraction)	**Low**

7. Molecular Mass and Formula Unit Mass

- **Molecular mass** (covalent compounds) = sum of atomic masses of all atoms in the molecule. Unit: **u**.
  H_2O = (1×2) + 16 = **18 u** | CO_2 = 12 + (16×2) = **44 u**
- **Formula unit** = simplest whole-number ratio of ions in an ionic compound (ionic compounds do not form molecules, so we use **formula unit mass**).
  Na_2O = (23×2) + 16 = **62 u** | $Ca(NO_3)_2$ = 40 + 2×(14 + 16×3) = **164 u**

8. Quick Revision Points

- Mass is conserved in every chemical reaction (Lavoisier); a compound has a fixed mass ratio of elements (Proust).
- Covalent bond = sharing (single, double, triple); ionic bond = transfer, giving cations and anions held by electrostatic attraction.
- Metals form cations; non-metals form anions. Ionic compounds are high-melting, conduct only when dissolved/molten, and form crystals.
- Criss-cross valencies, then reduce to simplest ratio; use brackets for repeated polyatomic ions.
- **Common mistakes:** writing 'monooxide' (it is *monoxide*); writing Mg_2O_2 instead of MgO; $AlOH_3$ instead of $Al(OH)_3$; calling ionic compounds 'molecules'.